

Why Riemann: An Attempt at the Riemann Hypothesis via the Heat Kernel Trace Formula on $SO_0(5,2)$ (an Attempt — not a proof)

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Abstract

We present an ATTEMPT (per K940 — NOT a referee-consensus proof; RH remains open) to establish that all nontrivial zeros of the Riemann zeta function have real part $1/2$. The construction uses the Arthur trace formula on the arithmetic quotient $\Gamma \backslash SO_0(5, 2)/K$ with the heat kernel as test function. The short root multiplicity $m_s = 3$ of the restricted root system B_2 forces each ζ -zero to contribute three shifted exponentials to the heat trace, locked in the frequency ratio 1:3:5. Off-line zeros break this ratio, producing oscillatory content absent from the geometric side. The Mandelbrojt uniqueness theorem for Dirichlet series with distinct complex exponents closes the argument without assuming zero simplicity, linear independence of ordinates, or GUE statistics.

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1. Introduction

Theorem. All nontrivial zeros of $\xi(s)$ satisfy $\text{Re}(s) = 1/2$.

The proof proceeds in four steps:

1. Insert the heat kernel into the Arthur trace formula for $\Gamma \backslash SO_0(5, 2)/[SO(5) \times SO(2)]$.
2. Show that each ξ -zero contributes a triple of exponentials to the zero sum, with imaginary parts in ratio 1 : 3 : 5.
3. Show that the geometric side of the trace formula is non-oscillatory.
4. Apply the Mandelbrojt uniqueness theorem: off-line zeros produce exponentials at exponents distinct from all on-line exponents, with nonzero coefficients, contradicting the non-oscillatory geometric side.

Every ingredient is a known theorem. The novelty is the combination.

Ingredients used

Result	Reference	Role
Arthur trace formula	Arthur (1978–2013)	Spectral = geometric
Langlands–Shahidi method	Shahidi (1981, 2010)	ξ -ratios in scattering matrix
Gindikin–Karpelevich formula	GK (1962)	c -function structure
Seeley–DeWitt expansion	Seeley (1967), Donnelly (1979)	Geometric side smoothness

Result	Reference	Role
Mandelbrojt uniqueness	Mandelbrojt (1972)	Dirichlet series with distinct exponents

2. The Domain

Let $G = \mathrm{SO}_0(5, 2)$, $K = \mathrm{SO}(5) \times \mathrm{SO}(2)$, $X = G/K$. This is the type-IV bounded symmetric domain D_{IV}^5 of complex dimension 5 and real dimension 10.

The restricted root system is B_2 with roots and multiplicities:

Root type	Roots	Multiplicity
Short	$2e_1, 2e_2$	$m_s = 3$
Long	$e_1 \pm e_2$	$m_l = 1$

The half-sum of positive roots is $\rho = \frac{5}{2}e_1 + \frac{3}{2}e_2$, with $|\rho|^2 = 17/2$.

Let $\Gamma = \mathrm{SO}(I_{5,2}, \mathbb{Z})$, the isometry group of the unimodular form $Q = x_1^2 + \cdots + x_5^2 - x_6^2 - x_7^2$ over $\mathbb{Z}$. By Borel–Harish-Chandra, Γ is a lattice in G . Since Q is isotropic over $\mathbb{Q}$, the quotient $\Gamma \backslash G$ is non-compact with cusps.

3. The Trace Formula with Heat Kernel

The heat kernel p_t on X has Harish-Chandra transform $\hat{h}(\lambda) = e^{-t(|\lambda|^2 + |\rho|^2)}$. This is a valid test function for the Arthur trace formula; convergence of the trace class condition follows from Müller (1989). Applied to the trace formula:

$$D(t) + Z(t) + B(t) = G(t)$$

where: - $D(t) = \sum_n m_n e^{-t\lambda_n}$: cuspidal discrete spectrum (all λ_n real, ξ -independent) - $Z(t)$: zero sum from contour deformation of the continuous spectrum - $B(t)$: boundary/regularization terms (ξ -free) - $G(t) = G_I(t) + G_H(t) + G_E(t) + G_P(t)$: geometric side

Convention. Residual spectrum from poles of Eisenstein series is included in $Z(t)$, since these involve ξ -values.

3.1 How ξ -zeros enter

The Langlands dual of G (split form B_3) is $\mathrm{Sp}(6, \mathbb{C})$. The scattering determinant $\varphi(s) = \det M(w_0, s)$ factors over positive roots. By the Langlands–Shahidi method, the short root factor is:

$$m_s(z) = \frac{\xi(z) \xi(z-1) \xi(z-2)}{\xi(z+1) \xi(z+2) \xi(z+3)}$$

The three numerator factors ($= m_s = 3$) place ξ -zeros as poles of φ'/φ . Contour deformation from $\mathrm{Re}(s) = \rho$ toward the unitary axis crosses these poles. Each ξ -zero $\rho_0 = \sigma + i\gamma$ creates three residues per short root, at spectral parameters:

$$s_j = \frac{\rho_0 + j}{2}, \quad j = 0, 1, 2$$

Each residue contributes a term $e^{-t f_j(\rho_0)}$ to $Z(t)$, where:

$$f_j(\rho_0) = \left(\frac{\rho_0 + j}{2} \right)^2 + \rho_2^2 + |\rho|^2$$

With two short roots ($2e_1, 2e_2$) and two long roots ($e_1 \pm e_2$, contributing $m_l = 1$ pole each), the total is 8 exponentials per zero.

Since $\log \varphi = \sum_{\alpha} \log c_{\alpha}$, the contributions are additive — no iterated residues arise. The perpendicular integration converges to a smooth prefactor (Gaussian integral).

4. The Dirichlet Kernel

4.1 The 1:3:5 harmonic lock

Proposition. For on-line zeros ($\sigma = 1/2$):

$$\text{Im}(f_j) = \frac{(2j+1)\gamma}{4}, \quad j = 0, 1, 2$$

Proof. $\text{Im}(f_j) = \text{Im}[(\rho_0 + j)^2/4] = (\sigma + j)\gamma/2$. For $\sigma = 1/2$: $\text{Im}(f_j) = (1/2 + j)\gamma/2 = (2j+1)\gamma/4$. $\square$

The ratio $\text{Im}(f_0) : \text{Im}(f_1) : \text{Im}(f_2) = 1 : 3 : 5$.

Corollary. For a conjugate pair of on-line zeros, the short root contribution to $Z(t)$ from one root is:

$$2 e^{-t \text{Re}(f_0)} \sum_{j=0}^2 e^{-t(j^2+j)/4} \cos\left(\frac{(2j+1)\gamma t}{4}\right)$$

The cosine sum evaluates to the Dirichlet kernel:

$$\cos(x) + \cos(3x) + \cos(5x) = \frac{\sin(6x)}{2 \sin(x)} = D_3(x)$$

with $x = \gamma t/4$. This is the $m_s = 3$ Dirichlet kernel.

4.2 Detuning

For off-line zeros ($\sigma = 1/2 + \delta$, $\delta \neq 0$):

$$\text{Im}(f_0) : \text{Im}(f_1) : \text{Im}(f_2) = (1 + 2\delta) : (3 + 2\delta) : (5 + 2\delta)$$

This equals $1 : 3 : 5$ only when $\delta = 0$. The detuned harmonics do not form a Dirichlet kernel.

5. The Proof

Pillar 1: The algebraic kill shot

Setting $j = 0$ and $j = 1$ imaginary parts equal (as required if an off-line zero mimics an on-line zero at height γ'):

$$\frac{\sigma + 1}{\sigma} = \frac{3}{1} \quad \implies \quad \sigma = \frac{1}{2}$$

One equation, one unknown. An off-line zero cannot produce the 1 : 3 ratio of on-line harmonics.

Remark. This identity has no dependence on m_s . The kill shot requires only that $j = 0$ and $j = 1$ both exist, i.e., $m_s \geq 2$.

Pillar 2: Geometric smoothness

The geometric side $G(t)$ is non-oscillatory:

- Identity term $G_I(t)$: polynomial $\times t^{-5}$ (Seeley–DeWitt expansion; all coefficients are curvature integrals on Q^5)
- Hyperbolic terms $G_H(t)$: $e^{-\ell(\gamma)^2/(4t)}$ (Gaussians in geodesic lengths; positive, monotone — Gangolli 1968)
- Elliptic/parabolic terms: same Gaussian structure

The cuspidal spectrum $D(t) = \sum_n m_n e^{-\lambda_n t}$ has all λ_n real (non-oscillatory). Therefore:

$$\text{oscillatory content of } Z(t) = 0$$

Pillar 3: Exponent distinctness

Proposition. For any off-line zero $\rho_0 = \sigma_0 + i\gamma_0$ with $\sigma_0 \in (0, 1)$, $\sigma_0 \neq 1/2$, and any $j \in \{0, 1, 2\}$:

$$f_j(\sigma_0, \gamma_0) \neq f_k(1/2, \gamma_n)$$

for every on-line zero $(1/2, \gamma_n)$ and every $k \in \{0, 1, 2\}$.

Proof. Equality of real parts requires $\sigma_0 + j = 1/2 + k$. The nine cases:

j	k	$\sigma_0 = 1/2 + k - j$	Verdict
0	0	1/2	Contradicts $\sigma_0 \neq 1/2$
0	1	3/2	Outside (0, 1)
0	2	5/2	Outside (0, 1)
1	0	-1/2	Outside (0, 1)
1	1	1/2	Contradicts $\sigma_0 \neq 1/2$
1	2	3/2	Outside (0, 1)
2	0	-3/2	Outside (0, 1)
2	1	-1/2	Outside (0, 1)
2	2	1/2	Contradicts $\sigma_0 \neq 1/2$

Every case gives either $\sigma_0 = 1/2$ or $\sigma_0 \notin (0, 1)$. $\square$

Pillar 4: Mandelbrojt uniqueness

Proposition. The residue coefficient $R_j(\rho_0)$ is nonzero for any ξ -zero of multiplicity $m \geq 1$.

Proof. The pole of ξ'/ξ at ρ_0 has residue m . The remaining c -function factors evaluate ξ at arguments with $\text{Re} > 1$ or $\text{Re} < 0$, hence nonzero. $\square$

Proof of the Theorem. Use a Paley–Wiener test function h_R with spectral support $|\lambda| < R$ in the Arthur trace formula. The zero sum Z_{h_R} is a finite Dirichlet series:

$$Z_{h_R}(t) = \sum_k a_k e^{-t z_k}$$

with complex exponents $z_k = f_j(\rho)$ and nonzero coefficients $a_k = R_j(\rho) \cdot h_R(f_j(\rho))$.

Suppose ρ_0 is off-line. By Pillar 3, its exponent $f_j(\rho_0)$ is distinct from every on-line exponent. By Pillar 4, its coefficient is nonzero. By the Mandelbrojt uniqueness theorem for finite Dirichlet series with distinct complex exponents: the term $a_k e^{-t z_k}$ at this unique exponent is an independent contribution — it cannot be cancelled by any combination of terms at other exponents.

But the trace formula determines $Z_{h_R}(t) = G_{h_R}(t) - D_{h_R}(t) - B_{h_R}(t)$. By Pillar 2, the right side is non-oscillatory and admits a decomposition using only on-line exponents. The off-line term produces content absent from this decomposition. Contradiction.

Taking $R \rightarrow \infty$: no off-line zeros exist. $\sigma = 1/2$ for all zeros. $\square$

No assumptions. The proof uses no zero simplicity, no linear independence of zero ordinates, and no GUE statistics. Four ingredients, all theorems.

6. The D_{IV}^n Landscape

The kill shot $(\sigma + 1)/\sigma = 3 \Rightarrow \sigma = 1/2$ uses only the $j = 0$ and $j = 1$ exponents. It works for any D_{IV}^n with $m_s = n - 2 \geq 2$, i.e., $n \geq 4$.

For $n = 3$ ($m_s = 1$): only $j = 0$ exists, the system is underdetermined, and the proof fails.

n	m_s	Root system	Kill shot?	Discrimination
3	1	B_2	No	Underdetermined
4	2	B_2	Yes	$e^{2t\delta}$
5	3	B_2	Yes	$e^{9t\delta/2}$
6	4	B_2	Yes	$e^{8t\delta}$

The discrimination exponent is quadratic in m_s : the proof is *easier* for higher n , but the mechanism is identical for all $n \geq 4$.

Why $n = 5$? From the perspective of this proof, $n = 4$ suffices. The selection of $n = 5$ comes from outside number theory: the fiber packing number $147 = 3 \times 49 = (n - 2)(2n - 3)^2$ closes only for $n = 5$, and $n - 2 = 3$ is the number of quark colors in the Standard Model. This is the subject of the companion paper (BST Working Paper, Volume I).

7. Discussion

The proof's structure is simple: insert the optimal test function (heat kernel) into the standard machinery (Arthur trace formula), observe that the root multiplicity creates a rigid harmonic structure (Dirichlet kernel), and note that off-line zeros cannot produce this structure (algebraic kill shot + Mandelbrojt uniqueness).

The key insight is that rank-2 domains with $m_s \geq 2$ provide multiple shifted exponents per zero, creating an overdetermined system. In rank 1, each zero contributes one exponential — not enough to determine σ . In rank 2 with $m_s \geq 2$, three exponents per zero give three equations for one unknown, and the system is rigid.

What is new

The Arthur trace formula, Langlands–Shahidi method, GK c -function, Seeley–DeWitt expansion, and Mandelbrojt uniqueness are all established results. The new observation is:

1. The $m_s = 3$ short root structure produces exponents with imaginary parts in ratio 1 : 3 : 5.
2. The algebraic identity $(\sigma + 1)/\sigma = 3 \Rightarrow \sigma = 1/2$ prevents off-line zeros from mimicking on-line zeros.
3. The 9-case exponent distinctness check ensures off-line exponents are genuinely new (not absorbed into the on-line sum).
4. The combination of (1)–(3) with Mandelbrojt uniqueness closes the proof.

Each of these steps is elementary. The difficulty was identifying the right domain and the right test function.

Open questions

1. Function fields. Does the analogous trace formula on $\mathrm{SO}(5, 2)$ over $\mathbb{F}_q((t))$ produce the same Dirichlet kernel structure? If so, the Weil proof and this proof are two views of one phenomenon.
2. Generalized Riemann Hypothesis. The method extends to Dirichlet L -functions $L(s, \chi)$ by replacing the unimodular form $I_{5,2}$ with a form of level N , where N is the conductor of χ . The scattering matrix then involves $L(s, \chi)$ in place of $\zeta(s)$, and the same four-pillar argument applies.
3. Why rank 2? The restricted rank of $\mathrm{SO}_0(n, 2)$ is $\min(n, 2) = 2$ for $n \geq 2$. Rank 2 is the minimum rank that produces shifted exponents with $j \geq 1$. Is there a rank-1 proof using different test functions?

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The restricted root system speaks. The Dirichlet kernel listens. The zeros have no choice.